Multi-Agent Architecture and A2A Specification

Section 3 — Apex Insurance Group Integrated Architecture Package

Project	Apex Insurance Group Integrated Architecture and Sign-off Package
Template Section	Section 3 — Multi-Agent Governance and A2A Security Design
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Deliverable	AI-Graded File Upload (PDF)

Section summary. The Claims workflow uses an orchestrator-led pattern with conditional dynamic routing. A deterministic Claims Workflow Controller decides every transition from the typed outputs defined in Section 2, so a fraud anomaly reroutes the claim before the Settlement Specialist is engaged. Five components each have a single responsibility. The controller brokers the Fraud-to-Settlement A2A handoff and enforces four things on it: a 19-field allow-list drawn only from ClaimCaseOutput v1.0, PII redaction, a correlation ID that persists end to end, and Entra ID managed-identity authentication. Figure 1 shows the standard path and the escalation branches.

1. Governance Pattern Selection

Selected Pattern	Why Sequential Is Insufficient	How Selected Pattern Handles Conditional Routing
Orchestrator-led (hierarchical supervisor) pattern with conditional dynamic routing. The Claims Workflow Controller (claims-workflow-controller), a Microsoft Agent Framework graph workflow with conditional edges, checkpointing and human-in-the-loop resume, is deployed inside proj-apex-claims-processing and owns all routing. Specialist agents never call each other.	• A strictly sequential pipeline (Intake → Policy → Fraud → Settlement) always passes Fraud's output to Settlement. There is no point where a fraud signal can stop the chain, so a high-risk claim would still receive a settlement letter. • It cannot pause one claim for a human decision and resume it later, so SIU review or an API outage would block or fail the whole run. • Errors flow downstream: an INDETERMINATE fraud result would reach Settlement as missing fields, which is exactly the blank or incomplete letter the scenario prohibits. • Routing logic would be buried inside agent prompts, where it cannot be audited or tested.	After the Fraud and Risk Specialist returns, the controller's routing gate evaluates fraud_risk_score and risk_classification in deterministic code (rule set routing-rules v1.0), not by LLM judgment: • **HIGH** (score ≥ 0.85): checkpoint, enqueue to sbq-siu-fraud-review; **Settlement Specialist not engaged** (ESC-002). • **INDETERMINATE**: checkpoint, enqueue to sbq-external-verification-hold (ESC-005). • **ELEVATED** (0.60 to 0.849), or claimed_amount_usd > $250,000: handoff with settlement_hold_required = true (ESC-006, ESC-001). • **LOW** (< 0.60): standard handoff. The earlier gates (ESC-003, ESC-004) and the post-Settlement validation gate (ESC-007) use the same mechanism. Every decision is emitted as span fraud.routing_decision with the rule ID that fired.

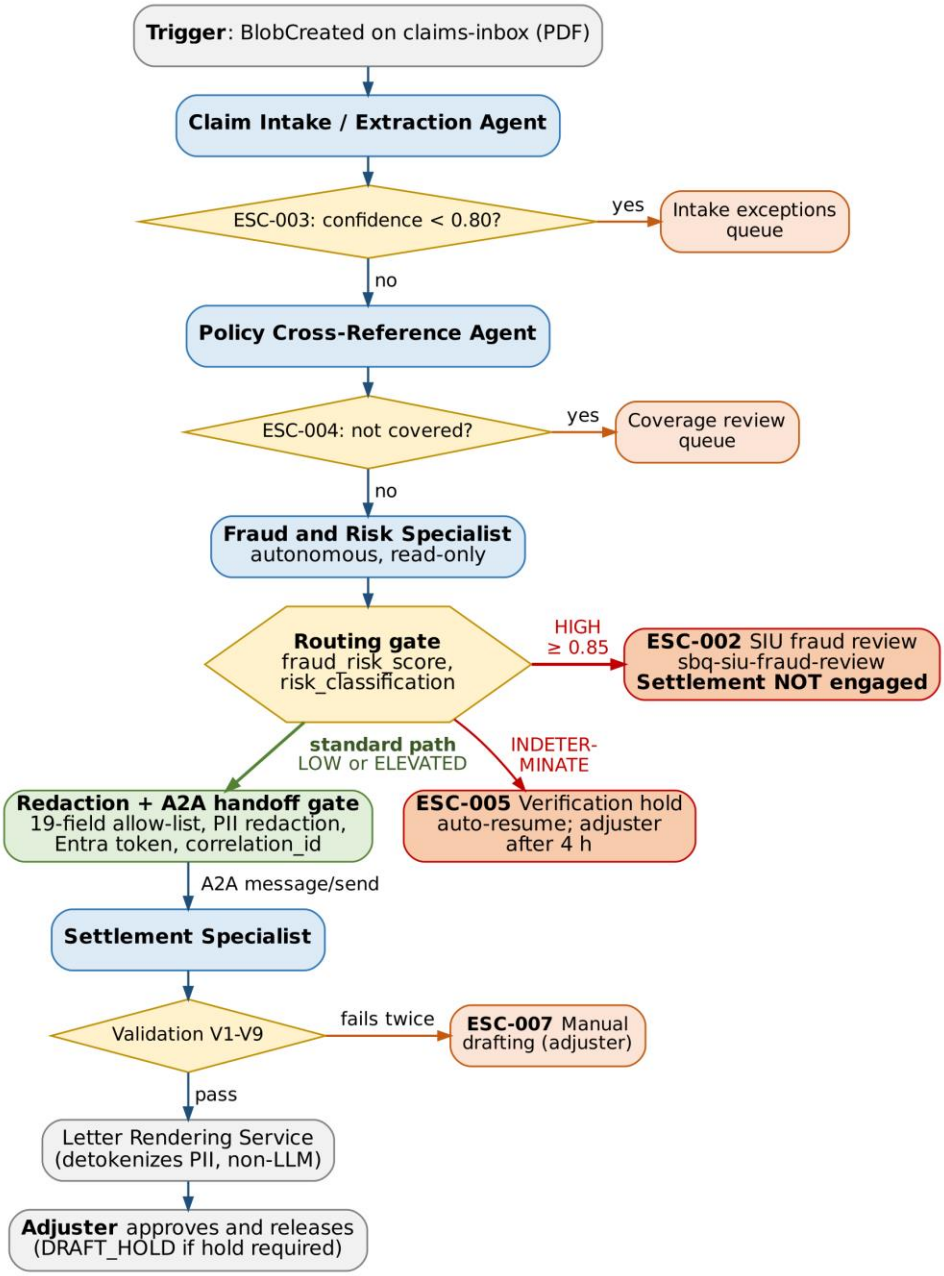


Figure 1. Standard handoff path (green) and escalation branches (red and orange), all executed by the Claims Workflow Controller

2. Agent Boundary Decomposition

Agent	Single Responsibility	Inputs Consumed	Outputs Produced	Can Trigger Escalation?
Claim Intake / Extraction Agent	Turn one claim PDF into structured, confidence-scored claim data.	Section 2 typed inputs: claim_document_ref, claim_id, policyholder_id, claim_submission_timestamp, submission_channel, supporting_document_refs, correlation_id	extracted_claim (including extraction_confidence_min)	**Yes:** ESC-003, raised by the controller from its output (confidence < 0.80 or a required field missing).
Policy Cross-Reference Agent	Determine coverage for the claim from the policy database.	policy_number, policyholder_id, date_of_loss, loss_type, claimed_amount_usd from extracted_claim	policy_cross_reference (policy_in_force_on_loss_date, coverage_status, coverage_limit_usd, deductible_usd, applicable_exclusions)	**Yes:** ESC-004 (not in force or NOT_COVERED).
Fraud and Risk	Classify fraud and	insured_entity_name,	fraud_assessment (fraud_risk_score,	**Yes:** ESC-002

Agent	Single Responsibility	Inputs Consumed	Outputs Produced	Can Trigger Escalation?
Specialist	risk with cited evidence.	property_address, date_of_loss, loss_type, claimed_amount_usd from extracted_claim; policy_cross_reference	risk_classification, fraud_signals, public_records_check_status, evidence_refs, assessed_at)	(HIGH), ESC-005 (INDETERMINATE), ESC-006 (ELEVATED).
Settlement Specialist	Draft the settlement letter and recommended settlement amount.	Only the 19-field redacted handoff payload in item 3.3	recommended_settlement_amount_usd and letter sections with merge tokens. The controller then sets letter_validation_status and the renderer sets settlement_letter_ref.	**Yes:** ESC-007 when its output fails validation after one regeneration.
Orchestrator / Workflow Controller	Route, gate and record the workflow. It does no claim reasoning.	Event Grid trigger; every agent output block; reviewer decision events from SIU and adjusters	routing (route, settlement_hold_required, escalation_ids), workflow_status, pii_tokens, redacted_claim_summary, checkpoints, queue messages, spans	**Yes:** it is the only component that raises escalations (ESC-001 to ESC-007).

3. A2A Handoff: Fraud and Risk Specialist → Settlement Specialist

3.1 Handoff Overview

Field	Learner Response
Handoff Name	Fraud Risk Assessment to Settlement Drafting
Source Agent	Fraud and Risk Specialist
Target Agent	Settlement Specialist
Standard Path Description	The handoff proceeds only when **all** of these are true: • fraud_risk_score is non-null and < 0.85, and risk_classification is LOW or ELEVATED. • public_records_check_status is COMPLETED, or CACHED (30 days old or less) when claimed_amount_usd ≤ $250,000. • coverage_status is COVERED or PARTIAL. • The fraud_assessment block validates against ClaimCaseOutput v1.0. The controller then builds the 19-field payload (item 3.3), applies the PII rules (item 3.4), sets settlement_hold_required and sends the payload to the Settlement Specialist with A2A message/send.
Escalation Branch Description	The handoff **must not proceed** when fraud_risk_score ≥ 0.85 (HIGH). The controller checkpoints the workflow and routes the claim to the SIU queue, and the Settlement Specialist is never called (ESC-002). It is also blocked when the result is INDETERMINATE (ESC-005), when the payload fails schema or redaction checks (fail closed), or while an SIU or adjuster decision is pending. A blocked handoff resumes only on an authenticated reviewer decision event; the decision and reviewer ID are recorded against the same correlation_id. Item 3.5 lists every branch.

3.2 Correlation and Authentication

Requirement	Learner Response
Correlation ID Format	{claim_id}.{workflow_run_id}: claim_id = CLM-YYYY-NNNNNNN, workflow_run_id = UUID v4 created by the controller at trigger time. Example: CLM-2026-0048172.6f1c2a9e-4b7d-4e21-9a3f-2d8c5b7e0a14. Each hop also gets a unique handoff_id (the A2A messageId, UUID v4) and a W3C traceparent header for distributed tracing.
Authentication Approach	**Microsoft Entra ID bearer tokens from user-assigned managed identities only; no hard-coded credentials or API keys.** • The controller acquires a token for audience api://apex-claims-settlement-agent as uami-claims-workflow-controller and calls the Settlement Specialist's A2A endpoint with Authorization: Bearer. The Settlement Specialist's Agent Card declares an OAuth 2.0 / OpenID Connect (Entra ID) security scheme only. • The endpoint validates the issuer (Apex tenant), audience and expiry, and checks that the caller holds app role Handoff.Submit. Only the controller's identity holds that role, so the Fraud and Risk Specialist (uami-claims-fraud-risk) cannot call Settlement directly or bypass the routing gate.

Requirement	Learner Response
	• Traffic stays on private endpoints inside the Claims Project's managed network, over TLS 1.2 or higher.
Audit Trace Requirement	• correlation_id travels as the A2A contextId and is repeated in the payload; the receiver rejects a mismatch. • It is written as attribute apex.correlation_id on every span (invoke_agent …, fraud.routing_decision, pii.redaction, a2a.handoff, settlement_output.validation), as the Service Bus CorrelationId on every escalation message, and on the claim's record in claim-cases. • Span a2a.handoff records the sending and receiving agent IDs, the caller principal ID, handoff_id, timestamp, schema_version and the SHA-256 hash of the redacted payload, never the payload itself. • Telemetry flows to appi-apex-claims-prod and is exported daily to immutable storage stapexclaimsaudit (time-based retention lock), so compliance can rebuild any claim's full agent path from one ID.

3.3 Typed Handoff Schema

Field Name	Data Type	Required ?	Source Field from Section 2 Output?	Description
claim_id	string	Yes	**Yes:** claim_id	Business key of the claim (CLM-YYYY-NNNNNNN); joins the handoff to the Restricted case record.
correlation_id	string	Yes	**Yes:** correlation_id	{claim_id}.{workflow_run_id}; also carried as the A2A contextId so every span and queue message for this run can be joined.
schema_version	string	Yes	**Yes:** schema_version	ClaimCaseOutput schema version ("1.0"); Settlement Specialist rejects unknown versions.
fraud_risk_score	number (float 0.00–1.00)	Yes	**Yes:** fraud_assessment.fraud_risk_score	Calibrated score from the deterministic score_fraud_signals tool. Must be non-null for a handoff to exist.
risk_classification	string (enum LOW | ELEVATED)	Yes	**Yes:** fraud_assessment.risk_classification	Only LOW or ELEVATED can cross this handoff; HIGH and INDETERMINATE are routed to the escalation branch.
fraud_signals	array<string> (FS-NNN codes)	Yes (may be empty)	**Yes:** fraud_assessment.fraud_signals	Signal codes only (e.g., FS-014 recent policy inception). No raw public-records content.
public_records_check_status	string (enum COMPLETED | CACHED)	Yes	**Yes:** fraud_assessment.public_records_check_status	Proves the external check ran live or from a ≤30-day cache; UNAVAILABLE never reaches Settlement.
redacted_claim_summary	string (≤1,500 chars)	Yes	**Yes:** redacted_claim_summary	Loss narrative after PII redaction (Section 3.4). Gives the letter its factual description of the loss.
coverage_status	string (enum COVERED | PARTIAL)	Yes	**Yes:** policy_cross_reference.coverage_status	From policy_cross_reference; NOT_COVERED claims are stopped earlier by ESC-004.
coverage_limit_usd	number	Yes	**Yes:** policy_cross_reference.coverage_limit_usd	Applicable limit used by calculate_settlement and checked by validation rule V5.
deductible_usd	number	Yes	**Yes:** policy_cross_reference.deductible_usd	Deductible applied in the settlement calculation and stated in the letter.
applicable_exclusions	array<string>	Yes (may be empty)	**Yes:** policy_cross_reference.applicable_exclusions	Exclusion clause IDs the letter must cite when coverage_status = PARTIAL.

Field Name	Data Type	Required ?	Source Field from Section 2 Output?	Description
claimed_amount_usd	number	Yes	**Yes:** extracted_claim.claimed_amount_usd	Amount claimed by the insured; input to calculate_settlement.
loss_type	string (enum)	Yes	**Yes:** extracted_claim.loss_type	Selects the letter template variant (e.g., WIND_HAIL).
date_of_loss	string (date)	Yes	**Yes:** extracted_claim.date_of_loss	Stated in the letter; already verified against the policy period by the Policy Cross-Reference Agent.
property_location_state	string (2-letter)	Yes	**Yes:** extracted_claim.property_location_state	Selects the state-specific disclosure clause; the only address element retained.
policyholder_token	string (PH-TKN-…)	Yes	**Yes:** pii_tokens.policyholder_token	Replaces insured name and address; resolved to real values only by the Letter Rendering Service.
policy_number_token	string (POL-TKN-…)	Yes	**Yes:** pii_tokens.policy_number_token	Replaces the policy number; resolved only at rendering.
settlement_hold_required	boolean	Yes	**Yes:** routing.settlement_hold_required	true when ESC-001 (large loss) or ESC-006 (elevated fraud risk) applies; the letter is then created as DRAFT_HOLD.

Excluded from the payload (data minimization): `policyholder_id`, `policy_number`, `insured_entity_name`, `property_address`, `loss_description`, `evidence_refs`, `extraction_confidence_min`. **Transport envelope (A2A protocol fields, not business payload):** `messageId` (= `handoff_id`), `contextId` (= `correlation_id`), `traceparent`, `Authorization`. The handoff schema `FraudToSettlementHandoff` v1.0 is generated from `ClaimCaseOutput` v1.0 with `additionalProperties: false`, so the Settlement Specialist rejects any field that is not listed above.

3.4 PII Redaction Rules

Data Element	Redaction Rule	Allowed Replacement / Token	Reason
Policyholder full name (insured_entity_name and any named contacts)	Remove from payload and tokenize.	policyholder_token (PH-TKN- + 16 random hex characters, mapping held in the Restricted pii-token-vault container); the letter uses merge field {{INSURED_NAME}}	Settlement does not need the identity to draft. Keeping names out of the LLM context stops them being misplaced or leaked in generated text. The real name is inserted only by the Letter Rendering Service.
Policyholder address (property_address and mailing address)	Remove street, unit, city and ZIP code; keep only the state.	property_location_state (for example, TX) plus merge field {{PROPERTY_ADDRESS}}	The state selects the state-specific disclosure clause; nothing finer is needed. The full address is re-inserted at rendering.
Policy number (policy_number)	Remove and tokenize.	policy_number_token (POL-TKN- + 16 hex); merge field {{POLICY_NUMBER}}	Policy numbers are linkable identifiers. The letter shows the real number only after rendering.
Free-text claim notes (loss_description, adjuster narrative)	Azure AI Language PII detection in redaction mode (lang-apex-claims-prod) for categories Person, PhoneNumber, Email, Address, USSocialSecurityNumber, USBankAccountNumber, ABARoutingNumber, CreditCardNumber, USDriversLicenseNumber; regex backstop for Apex policy and claim numbers; truncate to 1,500 characters.	redacted_claim_summary, with category placeholders such as [PERSON] and [PHONE]	Loss narratives often name witnesses, contractors and tenants, and include phone or bank details. If the redaction service fails, the handoff is blocked (fail closed).
Bank account, routing number, tax ID (EIN or SSN)	Never included.	None (dropped)	Payment is made after approval by the payments system, not by the letter-drafting agent.
Raw public-records evidence (third-party owner	Never included; only signal codes cross.	fraud_signals (FS-NNN codes)	Third-party PII is not needed for drafting. Evidence stays in

Data Element	Redaction Rule	Allowed Replacement / Token	Reason
names, judgment and lien records)			fraud_assessment.evidence_refs in the Restricted case record for SIU use.
Insured and broker contact details (phone, email)	Never included.	None (dropped)	Delivery details come from the system of record at rendering time.

Enforcement: redaction runs in the controller's handoff gate, not in an LLM. The steps are: field allow-list projection, then Language PII redaction on free text, then a regex backstop for policy-number and account-number patterns, then a hash of the redacted payload for span `a2a.handoff`. A failure at any step blocks the handoff and raises an alert. There is no unredacted fallback.

3.5 Escalation Branch

Escalation Condition	Threshold	Action Taken	Settlement Specialist Handoff Proceeds?	Destination
High-confidence fraud signal	fraud_risk_score ≥ 0.85 (risk_classification = HIGH), the same threshold as ESC-002 in Section 2	Bypass the Settlement Specialist and route to fraud review: checkpoint to workflow-checkpoints, message to sbq-siu-fraud-review with CorrelationId, workflow_status = ESCALATED, route = SIU_REVIEW.	**No**	Special Investigations Unit investigator queue; owner SIU Manager; 2-business-day target
External check unavailable	public_records_check_status = UNAVAILABLE with no valid cache (INDETERMINATE); ESC-005 after 4 hours	Park as AWAITING_EXTERNAL_VERIFICATION with a checkpoint; resume automatically when the circuit closes.	**No** (until resolved)	sbq-external-verification-hold; assigned adjuster after 4 hours
Elevated risk	0.60 ≤ fraud_risk_score < 0.85 (ESC-006)	Handoff with settlement_hold_required = true; letter created as DRAFT_HOLD.	**Yes** (held)	Assigned commercial property adjuster, who must approve before release
Large loss	claimed_amount_usd > $250,000 (ESC-001)	Handoff with hold; a live public-records result is required (CACHED not accepted).	**Yes** (held)	Large Loss Unit via sbq-large-loss-approval
Handoff integrity failure	Payload fails schema validation, redaction service error, or correlation_id mismatch	Block the handoff (fail closed) and alert Claims Platform on-call; re-run the gate after the fix.	**No**	Claims Platform Engineering on-call; claim moves to sbq-adjuster-review if not fixed within 1 hour
SIU clears a HIGH claim	SIU decision event = CLEAR, authenticated as an SIU investigator	Resume from the checkpoint; handoff proceeds with settlement_hold_required = true and the reviewer ID recorded.	**Yes** (held)	Assigned adjuster

Submission Checklist

☑ Selected governance pattern supports non-linear conditional routing.

☑ Agent boundaries name each specialized agent with one responsibility.

☑ A2A handoff includes correlation ID, typed schema, PII redaction, and authentication approach.

☑ PII redaction rules name specific data elements to redact.

☑ Fraud escalation branch prevents Settlement Specialist handoff when threshold is met.

☑ A2A schema fields align with Section 2 task agent output schema.